PAPER_158 â□□ Hybrid MUGE Blending Model: g_hybrid = Î²Â·g_compressed + (1-Î²)Â·g_resonance

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Abstract

This paper introduces the **Hybrid MUGE Blending Model**, a continuous interpolation between the Compressed MUGE and Resonance MUGE gravity frameworks. The blending parameter Î² = exp(â□□B/B_crit) transitions smoothly from Î² â□□ 0 (pure resonance, magnetar regime) to Î² â□□ 1 (pure compressed Newtonian+, weak-field regime). This unified hybrid is the **first algebraic bridge** between the Newtonian limit and the full resonance superconductive regime within the UQFF framework, extending Â§2.2 PAPER_155 which proved Newtonian emergence only in the fTRZâ□□0 limit.

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1. Motivation

PAPER_146 (Â§2.2) established the 12-term resonance MUGE master equation. PAPER_090 (Â§1.12) established the 9-term compressed MUGE. Previously, no **analytical bridge** existed for intermediate regimes where B is a significant fraction of B_crit (e.g., solar magnetosphere, white dwarfs, neutron star crusts).

The blending model provides: - Continuous function of B/B_crit across the full magnetic field parameter space - Smooth limiting behavior at both extremes - Single equation for all astrophysical environments

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2. Hybrid Blending Equation

$$g_{hybrid}(r,t) = \beta \cdot g_{comp}(r,t) + (1-\beta) \cdot g_{res}(r,t)$$

where

$$\boxed{\beta = e^{-B/B_{crit}}}$$

- g_{comp} = compressed MUGE gravity (PAPER_090, 9-term Newtonian+corrections)
- g_{res} = resonance MUGE gravity (PAPER_146/159, 12/13-term superconductive resonance)
- B = local magnetic field strength [T]
- B_{crit} = critical quantum field (4.4Ã□10Â¹Â³ T for magnetars)

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3. Limiting Cases

3.1 Magnetar Regime (B/B_crit â□□ 1)

$$\beta = e^{-1} \approx 0.368 \quad \Rightarrow \quad g_{hybrid} \approx 0.37\, g_{comp} + 0.63\, g_{res}$$

For SGR 1745-2900 (B = 3Ã□10Â¹Â¹ T, B_crit â□□ 4.4Ã□10Â¹Â³ T):

$$\beta_{SGR} = e^{-3\times10^{11}/4.4\times10^{13}} \approx 0.9933 \approx 1$$

â g_hybrid â g_comp for SGR 1745 (compressed dominant near-critical but below)

3.2 Solar Regime (B « B_crit)

For Sun (B_s = 10â»â´ T):

$$\beta_{Sun} = e^{-10^{-4}/4.4\times10^{13}} \approx 1.000000$$

â g_hybrid â g_comp (pure Newtonian+ regime)

3.3 Neutron Star Surface (B ~ 10Â¹Â² T)

$$\beta_{NS} = e^{-10^{12}/4.4\times10^{13}} \approx 0.977$$

â g_hybrid â 0.977 g_comp + 0.023 g_res (dominantly compressed; 2.3% resonance correction)

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4. Python Implementation (CP3)

```python
import math

def compute_hybrid_muge(g_compressed: float, g_resonance: float,
                        B: float, B_crit: float = 4.4e13) -> float:
    """
    Hybrid MUGE blending: g_hybrid = beta * g_comp + (1-beta) * g_res
    where beta = exp(-B/B_crit).

    Args:
        g_compressed: Compressed MUGE gravity [m/sÂ²]
        g_resonance:  Resonance MUGE gravity [m/sÂ²]
        B:            Local magnetic field [T]
        B_crit:       Critical field (default 4.4e13 T, magnetar)

    Returns:
        g_hybrid:     Blended MUGE gravity [m/sÂ²]
    """
    if B_crit <= 0:
        raise ValueError("B_crit must be positive")
    beta = math.exp(-B / B_crit)
    return beta * g_compressed + (1.0 - beta) * g_resonance
```

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5. Validation â 7-System Blending Table

System	B [T]	Î²	g_comp [m/sÂ²]	g_res [m/sÂ²]	g_hybrid [m/sÂ²]
SGR 1745-2900	3Ã10Â¹Â¹	0.9933	1.783Ã10Â³â¹	1.655Ã10â´âµ	1.785Ã10â»â´+dom

System	B [T]	Î^2	g_comp [m/sÂ^2]	g_res [m/sÂ^2]	g_hybrid [m/sÂ^2]
Sagittarius A*	1Ã□10â□»â□µ	1.0	1.816Ã□10Â^{3}â□	1.256Ã□10Â^{1}â□	â□° g_comp
Tapestry	1Ã□10â□»â□´	1.0	2.989Ã□10Â^{3}Â^1	1.257Ã□10Â^{1}Â^1	Â^{2}â□□ g_comp
Westerlund 2	1Ã□10â□»â□´	1.0	2.989Ã□10Â^{3}Â^1	1.257Ã□10Â^{1}Â^1	Â^{2}â□□ g_comp
Pillars	1Ã□10â□»â□´	1.0	1.989Ã□10Â^{2}â□	1.256Ã□10Â^{1}â□	â□µ g_comp
Rings	1Ã□10â□»â□µ	1.0	2.989Ã□10Â^{3}Â^3	1.257Ã□10Â^{1}Â^1	Â^{3}â□□ g_comp
Student’s Guide	1Ã□10â□»Â^{1}â□°	1.0	2.000Ã□10â□´â□	1.257Ã□10Â^{1}â□	µâ□¶ g_comp

Note: For all non-magnetar systems, Î^2 â□□ 1 so g_hybrid â□□ g_comp. The blending becomes significant only for systems with B/B_crit > 0.01 (i.e., B > 4.4Ã□10Â^{1}Â^1 T).

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6. Theoretical Significance

The hybrid model provides: 1. **A unified equation** replacing case-by-case selection of compressed vs. resonance 2. **Continuous parameter space** enabling interpolation between observational regimes 3. **Automatic mode selection**: Î^2 encodes the magnetic suppression factor from PAPER_090 4. **Extends PAPER_155**: While PAPER_155 proves Newtonian emergence at fTRZâ□□0, this model handles intermediate B fields without requiring fTRZ â□□ 0

Connection to 4 UQFF Operational Modes (PAPER_064): - Compressed: Î^2 = 1 - Buoyant: Î^2 and Ubi terms active - Resonant: Î^2 = 0 - Superconductive: Î^2 = 0, fTRZ active

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Status: â□□ Complete | **CP Stage:** CP3 (new `compute_hybrid_muge()` function) **Supersedes:** N/A (new model) | **Related:** PAPER_090 (compressed), PAPER_146 (12-term resonance), PAPER_155 (Newtonian limit), PAPER_064 (4 modes)

UQFF computed: MUGE buoyancy ratio U_bi/F_U = [SSq]×?×r^2/GM = 5.7e-1×5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s^2 at r_ISCO.